

PROJECTIVE MODELS FOR HILBERT SQUARES OF K3 SURFACES

JIEAO SONG

ABSTRACT. Geometric descriptions for locally complete families of projective hyperkähler manifolds are only known in very few cases. We describe the projective geometry of the Hilbert square of a K3 surface of genus 7 or 8, by making use of the Mukai model: in both cases, it can be realized as a degeneracy locus on an ambient homogeneous space, in a strikingly similar fashion. In consequence, we give explicit descriptions of its ideal and syzygies, and extract some new information on the locally complete families. This is a joint work with Ángel Ríos Ortiz and Andrés Rojas.

1. K3 SURFACES

Definition 1.1. A *K3 surface* is a compact complex surface S with trivial canonical bundle $\omega_S \simeq \mathcal{O}_S$ and regular (that is, $h^1(\mathcal{O}_S) = 0$).

Having trivial canonical bundle also means that S admits a unique symplectic form: $H^0(S, \Omega_S^2) = \mathbb{C}\omega$.

1.1. Projective models. Let (S, L) be a projective K3 surface with L very ample, so we have an embedding

$$\varphi_L: S \hookrightarrow \mathbb{P}^g.$$

The number g is called the *genus* of S , since a general curve C in the linear system $|L|$ is a canonical curve of genus g . We have $\deg_L S = 2g - 2$.

Concerning the ideal of S in $\mathbb{P}^g$, we have the following classical result.

Theorem 1.2 (Saint-Donat). *Let S be a projective K3 surface.*

- *S is projectively normal.*
- *When $g \geq 5$ and S is general, the ideal of S is generated by quadrics.*

More concretely, Mukai has described explicitly the projective geometry of a general K3 surface in low degrees through a series of work. These are now known as the *Mukai models*.

Theorem 1.3. *Let (S, L) be a general K3 surface of genus g .*

- *$g = 2$: $\varphi_L: S \xrightarrow{2:1} \mathbb{P}^2$ is a double cover branched along a smooth sextic.¹*
- *$g \in \{3, 4, 5\}$: S is a $(4), (2, 3), (2, 2, 2)$ -complete intersection in $\mathbb{P}^3, \mathbb{P}^4, \mathbb{P}^5$, respectively.*
- *$g = 6$: S is a $(1, 1, 1, 2)$ -complete intersection in $\text{Gr}(2, 5) \subset \mathbb{P}^9$.*
- *$g = 7$: S is a $\mathbb{P}^7$ -linear section of $\text{OGr}_+(5, 10) \subset \mathbb{P}^{15}$.*
- *$g = 8$: S is a $\mathbb{P}^8$ -linear section of $\text{Gr}(2, 6) \subset \mathbb{P}^{14}$.*
- *$g = 9$: S is a $\mathbb{P}^9$ -linear section of $\text{IGr}(3, 6) \subset \mathbb{P}^{13}$.*
- *$g = 10$: S is a $\mathbb{P}^{10}$ -linear section of $G_2 \subset \mathbb{P}^{13}$.*

Date: July 10, 2025.

¹Note that L is not very ample. Still, this case fits nicely into the picture since genus 2 curves are hyperelliptic.

- $g \in \{11, 12, 13, 16, 18, 20\}$: *explicit descriptions*.

1.2. Syzygies. For a projective variety $(X, L) \subset \mathbf{P}^n$, apart from the projective normality and the generators of the ideal, most information about the embedding is encoded in its *syzygies*, which we now recall. We assume that X is embedded via a complete linear system $\varphi_L: X \hookrightarrow \mathbf{P}^n$.

Write $S := \mathbf{C}[x_0, \dots, x_n] = \bigoplus_{d \geq 0} S_d$ the polynomial ring, and $R := \bigoplus_{d \geq 0} H^0(X, L^d)$ the *ring of sections* of X .² By Hilbert's syzygy theorem, there exists a unique (up to isomorphism) minimal graded free resolution

$$\dots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow R \longrightarrow 0,$$

where each term F_i is a graded free S -module that can be written as

$$F_i = \bigoplus_j S(-i-j)^{b_{i,j}}.$$

We can place the numbers $b_{i,j}$, which are called the *Betti numbers*, in the following fashion to form the *Betti diagram* of (X, L) . We note that they satisfy upper semicontinuity.

	0	1	2	3	...
0	1	.	.	.	...
1	.	$b_{1,1}$	$b_{2,1}$	$b_{3,1}$	...
2	$b_{0,2}$	$b_{1,2}$	$b_{2,2}$	$b_{3,2}$	...
3	$b_{0,3}$	$b_{1,3}$	$b_{2,3}$	$b_{3,3}$	...

The Betti diagram offers a very convenient way to visually understand the embedding of X :

- The numbers $b_{0,d}$ in the column $i = 0$ describe, inductively, the cokernel of the natural map $S \rightarrow R$: if X is projectively normal, then they are all zero; otherwise, $b_{0,2}$ is the number of extra sections of $H^0(X, L^2)$ that are not induced by quadrics on $\mathbf{P}^n$, and $b_{0,3}$ is the number of extra sections of $H^0(X, L^3)$ that are not induced by cubics and the previous $b_{0,2}$ sections, etc.
- If X is projectively normal, then the numbers $b_{1,d}$ in the column $i = 1$ are precisely the numbers of generators in degree d of the ideal I of X .
- The numbers $b_{2,d}$ in the column $i = 2$ describe the *syzygies* or linear relations with polynomial coefficients among the generators of the ideal.
- The column $i + 1$ describes the syzygies among the generators in the column i .

Let us look at the Betti diagrams of general K3 surfaces as an example.

Example 1.4.

- For $g \in \{3, 4, 5\}$, since S is a complete intersection, we know that the minimal resolution can be given by the Koszul complex. Specifically, this means we have the following Betti diagrams, where we can easily read the number of equations in each degree and their syzygies

	0	1		0	1	2		0	1	2	3
0	1	.	0	1	.	.	0	1	.	.	.
1	.	.	1	.	1	.	1	.	3	.	.
2	.	.	2	.	1	.	2	.	.	3	.
3	.	1	3	.	.	1	3	.	.	.	1

²The name *ring of power* would also seem appropriate. In any case, when X is projectively normal, R is isomorphic to the *coordinate ring* S/I_X .

- When $g = 6$, we have the following Betti diagram for a general K3 surface

	0	1	2	3	4
0	1	.	.	.	.
1	.	6	5	.	.
2	.	.	5	6	.
3	.	.	.	.	1

We can interpret some of the numbers here: $b_{1,1} = 6 = 5 + 1$ since we have 5 Plücker quadrics and we are picking one more quadric for S ; $b_{2,1} = 5$ is given by the 5 linear syzygies for the Plücker quadrics.

- In genus $g \in \{7, 8, 9, 10\}$, the Betti diagram of a general S will be the same as that of the ambient homogeneous variety. For example, when $g = 7$ we have

	0	1	2	3	4	5
0	1	.	.	.	.	.
1	.	10	16	.	.	.
2	.	.	.	16	10	.
3	.	.	.	.	.	1

With these examples, one can immediately spot a few patterns and rules that these Betti diagrams seem to follow.

- The last non-trivial row is always $j = 3$: this is the *Castelnuovo–Mumford regularity*.
- The last non-trivial column is always $i = g - 2$: this means the resolution stops at the $(g - 2)$ -th term, which gives the *projective dimension*. The graded Auslander–Buchsbaum formula then states that R is Cohen–Macaulay, in which case X is called *arithmetically Cohen–Macaulay*.
- There is a clear symmetry of the Betti diagram.

These three properties follow directly from the following.

Theorem 1.5 (Green’s duality theorem). *Let $(X, L) \subset \mathbf{P}^n$ be a smooth projective variety with trivial canonical bundle $\omega_X \simeq \mathcal{O}_X$. Then we have the symmetry*

$$(1) \quad b_{i,j} = b_{\text{codim}_X/\mathbf{P}^n - i, \dim X + 1 - j}$$

for all $j \geq \dim X + 1$. Moreover, if $h^1(\mathcal{O}_X) = 0$, then the symmetry (1) holds for all $j \geq \dim X$.

Finally, one notices that the row $j = 1$ always stops at the half of the diagram. This remarkable fact is the following.

Theorem 1.6 (Generic Green’s conjecture; Voisin). *For a general K3 surface of genus g , we have $b_{i,1} = 0$ for $i \geq \lfloor \frac{g}{2} \rfloor$.*

The generic Green’s conjecture can also be reformulated as the slogan that the syzygies for a general K3 surface are “as simple as possible”.

We also remark that a canonical curve $C \in |L|$ has the same Betti diagram as S , thus by semicontinuity this recovers the Betti diagram of a generic canonical curve, not necessarily lying on any K3 surfaces.

2. HYPERKÄHLER MANIFOLDS

Hyperkähler manifolds are the natural generalization of K3 surfaces in higher dimensions.

Definition 2.1. A compact Kähler manifold X is called *hyperkähler* if it is simply connected (thus $h^1(\mathcal{O}_X) = 0$) and admits a unique symplectic form: $H^0(X, \Omega_X^2) = \mathbb{C}\omega$.

Deformation theory and moduli spaces for hyperkähler manifolds are well understood: almost everything is encoded in the second cohomology group $H^2(X, \mathbb{Z})$, which is equipped with a quadratic form q called the *Beauville–Bogomolov–Fujiki form*.

Example 2.2. Following is the list of known deformation types of hyperkähler manifolds in dimension ≥ 4 .

- For a K3 surface S , its Hilbert scheme of n points, denoted by $S^{[n]}$, is a hyperkähler manifold of dimension $2n$. We have

$$H^2(S^{[n]}, \mathbb{Z}) = H^2(S, \mathbb{Z}) \oplus \mathbb{Z}\delta,$$

$$\text{NS}(S^{[n]}) = \text{NS}(S) \oplus \mathbb{Z}\delta,$$

where 2δ is the class of the divisor Δ parametrizing non-reduced subschemes.

Since $b_2(S^{[n]}) = b_2(S) + 1$, general deformations of $S^{[n]}$ are no longer Hilbert schemes.

We call them *hyperkähler manifolds of K3^[n]-type*.

- A similar construction for abelian surfaces yields *generalized Kummer varieties* and *hyperkähler manifolds of Kum_n-type*.
- Two sporadic examples constructed by O’Grady in dimension 6 and 10.

2.1. Projective models. For K3 surfaces (S, L) , we used the genus g , or equivalently, the degree $\deg_L S = 2g - 2$ to characterize the polarizations. In higher dimensions, however, there are at least two invariants for a polarization H on a hyperkähler manifold X .

- The *square* $q(H) \in 2\mathbb{Z}$;
- The *divisibility* $\text{div}(H)$ which is the generator for the subgroup $q(H, H^2(X, \mathbb{Z})) \triangleleft \mathbb{Z}$.

From now on we focus on hyperkähler fourfolds of K3^[2]-type. We denote by $\mathcal{M}_{2d}^{(\gamma)}$ the moduli space of polarized (X, H) with $q(H) = 2d$ and $\text{div}(H) = \gamma$.

Proposition 2.3. $\mathcal{M}_{2d}^{(\gamma)}$ is non-empty in the following cases

- Either $\gamma = 1$ and $2d > 0$ is any even integer;
- Or $\gamma = 2$ and $2d \equiv 6 \pmod{8}$.

In all cases, it is always irreducible of dimension 20.

One would like to describe the general members in each locally complete family, in the same vein as the Mukai models for K3 surfaces.

Example 2.4. For K3^[2]-type, only the following examples are known.

- $\mathcal{M}_2^{(1)}$: (O’Grady) H in this case is not very ample; $X \xrightarrow{2:1} Y_6 \subset \mathbb{P}^5$ are double covers of certain special sextics so they are called *double EPW sextics*.
- $\mathcal{M}_6^{(2)}$: (Beauville–Donagi) $X \subset \text{Gr}(2, 6) \subset \mathbb{P}^{14}$ is the Fano variety of lines $F_1(Y)$ of a smooth cubic fourfold $Y \subset \mathbb{P}^5$. This is the first example to be studied and the most famous / most well-understood case.
- $\mathcal{M}_{22}^{(2)}$: (Debarre–Voisin) $X \subset \text{Gr}(6, 10)$ is known as the *Debarre–Voisin fourfold*, constructed from an element $\sigma \in \bigwedge^3 V_{10}^\vee$.

- $\mathcal{M}_{38}^{(2)}$: (Iliev–Ranestad) X is the *variety of sum of powers* of a cubic fourfold Y .

Note in particular that nothing is known for the lowest degree ones $\mathcal{M}_4^{(1)}$ and $\mathcal{M}_6^{(1)}$.

2.2. Hilbert squares of K3 surfaces. Since we do not have the description of a general member, the best we can do is to study Hilbert squares $S^{[2]}$ and try to deduce from them properties that a general member may satisfy.

Let $(S, L) \subset \mathbf{P}^g$ be a projective K3 surface of genus g . Let us first consider the polarization $L - \delta$ on $S^{[2]}$: it is given by the following natural map

$$(2) \quad \begin{aligned} \varphi_{L-\delta}: S^{[2]} &\longrightarrow \mathrm{Gr}(2, g+1) \\ (x, y) &\longmapsto [\overline{xy}] \end{aligned}$$

where a subscheme is sent to the class of the line it spans. We have $\varphi_{L-\delta}^* \mathcal{O}(1) = L - \delta$. Note that $L - \delta$ is very ample if and only if L is *2-very ample* on S , by definition. We have the following result.

Proposition 2.5. *Let (S, L) be a K3 surface of genus g . Assuming moreover that $\mathrm{NS}(S) = \mathbf{Z}L$. Then*

- When $g \geq 5$, L is 2-very ample.
- When $g \geq 3$, $2L$ is 2-very ample.

Since being very ample is an open property, we get the following.

Corollary 2.6. *A general $(X, H) \in \mathcal{M}_{2d}^{(\gamma)}$ is very ample when $2d \geq 6$, for either divisibility 1 or 2.*

Thus for very ampleness, $\mathcal{M}_4^{(1)}$ remains the last unknown case.

Now we consider the polarization $L - 2\delta$ which has the following interpretation. Let (S, L) be a K3 surface of genus $g \geq 5$ with $\mathrm{NS}(S) = \mathbf{Z}L$. By Saint-Donat's result, the ideal of S is generated by quadrics. We consider the following map

$$(3) \quad \begin{aligned} \varphi_{L-2\delta}: S^{[2]} &\longrightarrow |I_S(2)|^\vee \\ (x, y) &\longmapsto \{Q \in I_S(2) \mid Q \supset \overline{xy}\} \end{aligned}$$

where a subscheme is mapped to the hyperplane of quadrics containing the line it spans; since S cannot contain the line by the assumption that $\mathrm{NS}(S) = \mathbf{Z}L$, the map is indeed everywhere defined. We have $\varphi_{L-2\delta}^* \mathcal{O}(1) = L - 2\delta$.

Our first contribution is the following.

Theorem 2.7. *Let (S, L) be a K3 surface of genus $g \geq 7$ with $\mathrm{NS}(S) = \mathbf{Z}L$. Then $(S^{[2]}, L - 2\delta)$ is very ample.³*

By taking $g = 7$ this settles very ampleness for the remaining case $\mathcal{M}_4^{(1)}$.

³The proof involves a detailed analysis of the rank loci in $|I_S(2)|$, and their relations with moduli spaces of stable sheaves on S .

3. PROJECTIVE GEOMETRY OF $(S^{[2]}, L - 2\delta)$

The goal is now to understand the projective geometry of $(S^{[2]}, L - 2\delta)$, in the two cases of $g \in \{7, 8\}$ (they are elements of $\mathcal{M}_4^{(1)}$ and $\mathcal{M}_6^{(1)}$, respectively), by making use of their Mukai models. Recall that

- When $g = 7$, a general S is a $\mathbf{P}^7$ -linear section of $\text{OGr}_+(5, 10) \subset \mathbf{P}^{15}$;
- When $g = 8$, a general S is a $\mathbf{P}^8$ -linear section of $\text{Gr}(2, 6) \subset \mathbf{P}^{14}$.

We focus on the case of $g = 7$; the case of $g = 8$ turns out to be extremely similar.

3.1. Orthogonal Grassmannian. First, we recall some basic properties of the orthogonal Grassmannian $\text{OGr}_+(5, 10)$. Let V_{10} be a 10-dimensional vector space equipped with a quadratic form $q \in \text{Sym}^2 V_{10}^\vee$. Note that the quadratic form provides us with an isomorphism $V_{10} \simeq V_{10}^\vee$. The orthogonal Grassmannian $\text{OGr}_+ := \text{OGr}_+(5, V_{10})$ is one of the two components parametrizing 5-dimensional (maximal) isotropic subspaces with respect to q . It has dimension 10 and admits the *spinor embedding*

$$\text{OGr}_+ \hookrightarrow \mathbf{P}^{15} = \mathbf{P}V_+,$$

where V_+ is one of the two spinor representations of $\text{SO}(10)$ of rank 16. We remark that twice the spinor embedding (that is, taking the Veronese of degree 2) gives us the Plücker embedding.

Proposition 3.1. *The ideal of OGr_+ is generated by quadrics, and there is a canonical identification $I_{\text{OGr}_+}(2) = V_{10}$.*

Since S is a linear section of OGr_+ , in consequence, we get also the identification $I_S(2) = I_{\text{OGr}_+}(2) = V_{10}$.

Lemma 3.2. *For $x, y \in S$, let U_x and U_y be the corresponding 5-dimensional isotropic subspaces. We have $\dim(U_x \cap U_y) = 1$, $\dim(U_x + U_y) = 9$.*

Thus we get two ways of mapping $S^{[2]}$ into a projective space.

Proposition 3.3.

- The map

$$\begin{aligned} \varphi: S^{[2]} &\longrightarrow \mathbf{P}V_{10}^\vee = |I_S(2)|^\vee \\ (x, y) &\longmapsto U_x + U_y \end{aligned}$$

coincides with the map $\varphi_{L-2\delta}$ in (3).⁴

- Composing φ with the isomorphism $\mathbf{P}V_{10}^\vee \simeq \mathbf{P}V_{10}$ induced by the quadratic form q , we get

$$\begin{aligned} \psi: S^{[2]} &\longrightarrow \mathbf{P}V_{10} = |I_S(2)| \\ (x, y) &\longmapsto U_x \cap U_y. \end{aligned}$$

The quadric $Q_{x,y}$ in the image is the unique quadric singular along the line $\overline{xy}$.

A first consequence is that, via the embedding ψ , $S^{[2]}$ is contained in the quadric $Q \subset \mathbf{P}V_{10}$, as $U_x \cap U_y$ must still be isotropic. Note that we have

$$h^0(\mathbf{P}^9, \mathcal{O}(2)) = 55 = h^0(S^{[2]}, \mathcal{O}(2)),$$

so $S^{[2]}$ lying on a quadric in particular means it is *not* projectively normal.

⁴For the proof, we check that, under the identification $V_{10} = I_S(2)$, U_x and U_y consist of quadrics containing the line $\overline{xy}$.

3.2. Degeneracy locus. Now we present a nice geometric construction to realize $S^{[2]}$ as a degeneracy locus in Q .

Consider the following incidence variety

$$\begin{array}{ccccc}
 & I_p & I := \{(p, [U]) \mid p \in \mathbf{P}(U)\} & & \\
 & \swarrow & \searrow \text{pr}_1 & & \searrow \text{pr}_2 \\
 p & & \mathbf{P}^9 \supset Q & & \text{OGr}_+ \subset \mathbf{P}^{15} = \mathbf{P}V_+
 \end{array}$$

For each $p \in Q$, the preimage $I_p := \text{pr}_1^{-1}(p)$ is a subvariety of OGr_+ of dimension 6. It is a classical result that I_p spans a $\mathbf{P}^7$ in $\mathbf{P}^{15}$, inside which it is a quadric hypersurface, and when globalizing this to the entire Q , we obtain an 8-dimensional vector bundle $\mathcal{S}_+$ known as the *spinor bundle*. Clearly, via

$$\mathcal{S}_+ \hookrightarrow V_+ \otimes \mathcal{O}_Q,$$

we can view $\mathcal{S}_+$ as a subbundle of the rank-16 trivial bundle $V_+ \otimes \mathcal{O}_Q$.

On the other hand, (S, L) is a $\mathbf{P}^7$ -linear section of OGr_+ . In other words, the linear span $H^0(S, L)^\vee$ is an 8-dimensional subspace of V_+ so it naturally admits an 8-dimensional quotient W

$$0 \longrightarrow H^0(S, L)^\vee \longrightarrow V_+ \longrightarrow W \longrightarrow 0.$$

Via composition, we get a map between two vector bundles of rank 8

$$\varphi: \mathcal{S}_+ \longrightarrow W \otimes \mathcal{O}_Q.$$

Proposition 3.4. *Via the embedding $\psi: S^{[2]} \hookrightarrow Q \subset \mathbf{P}^9$, $S^{[2]}$ coincides with the degeneracy locus $D_6(\varphi)$ scheme-theoretically.*

Proof. We first show that set-theoretically they are the same.

For $(x, y) \in S^{[2]}$ corresponding to U_x and U_y , ψ maps it to $p \in Q$ such that $\mathbf{C}p = U_x \cap U_y$. Hence x and y both lie in the preimage I_p ; $\mathbf{C}x + \mathbf{C}y$ gives us a 2-dimensional subspace of $\mathcal{S}_{+,p}$ that is mapped to 0 in W (since x and y lie on S). So $\text{rank } \varphi_p \leq 6$.

Conversely, if $p \in Q$ satisfies $\text{rank } \varphi_p = 6$, the 2-dimensional kernel gives us a $\mathbf{P}^1$ contained in both the linear span of I_p and the linear span of S . Since I_p is a quadric hypersurface, we obtain either a subscheme of length 2 or a line entirely contained in S . By the assumption that $\text{NS}(S) = \mathbf{Z}L$, we must be in the first case.

Also, the same argument shows that no $p \in Q$ can have $\text{rank } \varphi_p \leq 5$: otherwise we would get a conic or a $\mathbf{P}^2$ entirely contained in S .

To conclude that scheme-theoretically they are the same, we note that since $D_6(\varphi)$ is determinantal of the expected dimension, it must be Cohen–Macaulay and contains no embedded components. Thus it suffices to compute the degree using the Porteous’ formula and check that it coincides with the degree of $S^{[2]}$. $\square$

Since we can realize $S^{[2]}$ as a degeneracy locus, we now have the full machinery of determinantal ideals at our disposal. In the particular case of corank-2 square matrices, there is the four-term locally-free *Gulliksen–Negård* resolution

$$0 \longrightarrow F_4 \longrightarrow F_3 \longrightarrow F_2 \longrightarrow F_1 \longrightarrow \mathcal{I}_{S^{[2]}/Q} \longrightarrow 0.$$

Proposition 3.5. *Using the Gulliksen–Negård complex,⁵ we obtain*

- $S^{[2]}$ is d -normal for $d \geq 3$, and $I_{S^{[2]}}(2) = \mathbf{C}Q$ (so 2-normality only fails by one).
- $\mathcal{I}_{S^{[2]}/Q}$ is generated by 65 quartics.
- We can describe the full minimal resolution in terms of its Betti diagram, which is⁶

	0	1	2	3	4	5	6	7
0	1	.	.	.	.	.	.	.
1	.	<u>1</u>	.	.	.	.	.	.
2	<u>1</u>	10	.	.	.	.	.	.
3	.	20	126	190	130	<u>46</u>	10	1
4	.	.	.	.	<u>1</u>	.	.	.
5	.	.	.	.	.	1	.	.

The natural question now is what happens with a general deformation of $S^{[2]}$. With some computation of infinitesimal nature (namely, by counting first-order deformations of $S^{[2]}$ in Q and in $\mathbf{P}^9$), we are able to conclude that

Proposition 3.6. *A general deformation of $(S^{[2]}, L - 2\delta)$, that is, a general $(X, H) \in \mathcal{M}_4^{(1)}$, is not contained in any quadrics.*

Corollary 3.7. *A general $(X, H) \in \mathcal{M}_4^{(1)}$ is projectively normal with Betti diagram*

	0	1	2	3	4	5	6	7
0	1	.	.	.	.	.	.	.
1	.	.	.	.	.	.	.	.
2	.	10	.	.	.	.	.	.
3	.	20	126	190	130	45	10	1
4	.	.	.	.	.	.	.	.
5	.	.	.	.	.	1	.	.

Remark 3.8.

- This is probably the first example where the complete Betti diagram of a projective hyperkähler manifold (of dimension ≥ 4) has been computed. And in the spirit of the generic Green’s conjecture, it is indeed “as simple as possible”. This raises the interesting question of formulating and studying the generic Green’s conjecture in the context of hyperkähler manifolds.
- Note that the projective dimension is 7, larger than the codimension 5. This is expected, as $h^2(\mathcal{O}_X) = 1$ implies that X can never be embedded as an aCM variety.
- One can observe that the Betti diagram is the overlay of two parts: one symmetric part, and one extra “half-strand” of binomial coefficients in the row $j = 3$. Purely numerical computation suggests that in larger degrees, one might expect a similar decomposition.
- From the Betti diagram, we see that a general (X, H) is cut out by 10 cubics and 20 quartics. However, by computer experiments, one expects that X is already the

⁵Each F_i in the complex can be explicitly described in terms of homogeneous vector bundles on Q . Therefore, one can compute various cohomology groups on $S^{[2]}$ using the associated spectral sequence and the Borel–Weil–Bott theorem.

⁶The four terms that are marked with underline contain extra syzygies not expected from computing with the Hilbert function.

scheme-theoretical intersection of the 10 cubics; the quartics arise when we take the saturation of the ideal (that is, the corresponding affine scheme has an embedded component at the origin).

3.3. Genus 8. In the case of genus 8, we can repeat the construction with the incidence variety

$$\begin{array}{ccc} & I = \text{Flag}(2, 4, V_6) & \\ \text{pr}_1 \swarrow & & \searrow \text{pr}_2 \\ G := \text{Gr}(2, V_6^\vee) & & \text{Gr}(2, V_6) \subset \mathbf{P}(\wedge^2 V_6) \end{array}$$

The spans of fibers of pr_1 give us the rank-6 vector bundle $\wedge^2 \mathcal{Q}^\vee \hookrightarrow \wedge^2 V_6 \otimes \mathcal{O}_G$ on G , while the choice of the K3 surface S gives us a quotient $\wedge^2 V_6 \twoheadrightarrow W$ where W is of dimension 6. And via composition we get a morphism between two vector bundles of rank 6

$$\varphi: \wedge^2 \mathcal{Q}^\vee \longrightarrow W \otimes \mathcal{O}_G.$$

Proposition 3.9. $(S^{[2]}, L-2\delta)$ coincides with the degeneracy locus $D_4(\varphi)$ scheme-theoretically.

Remark 3.10. Note that the construction gives us two rank-2 vector bundles on $S^{[2]}$: namely, the kernel and the cokernel of the map φ . The kernel bundle can be seen as the *universal secant line* to the K3 surface S , so it has first Chern class $-(L - \delta)$, from which we deduce that the cokernel has first Chern class $2L - 5\delta$. On the other hand, the cokernel bundle can be seen as (dual of) the *universal line* contained in the Pfaffian cubic in $\mathbf{P}W^\vee$. So this construction can be linked nicely back to the Beauville–Donagi construction.

Corollary 3.11. *Using the degeneracy locus description, we can deduce the following.*

- $(S^{[2]}, L - 2\delta)$ is projectively normal (hence so is a general $(X, H) \in \mathcal{M}_6^{(1)}$).
- $\mathcal{I}_{S^{[2]}/G}$ is generated by 55 cubics (hence a general $(X, H) \in \mathcal{M}_6^{(1)}$ is cut out by 15 quadrics and at most 55 cubics).
- We can obtain part of the Betti diagram of $S^{[2]}$.

Remark 3.12. Again by computer experiments, one expects that a general (X, H) is already the scheme-theoretical intersection of the 15 quadrics. Moreover, there seems to exist a unique cubic hypersurface $\mathcal{C}$ such that the 15 quadrics are precisely the polars of $\mathcal{C}$, so X is its singular locus. This is very similar to the Coble hypersurfaces for abelian varieties. It would be very interesting to understand its geometry.